KIDS LEARNING ADVENTURE

Flutter Mobile App - Product, UX and Development Plan

Target audience: Pre-KG, LKG and UKG children (approximately ages 3-6). Platforms: Android and iOS. Release approach: offline-first MVP, with higher classes added in future versions.

Product vision: A highly visual, cartoon-based learning adventure that lets young children see, listen, touch, play, learn and earn friendly rewards.

1. Product concept and guiding principles

- Game-like experience rather than text-heavy lessons; a friendly animated animal mascot guides the child.
- Large touch targets, minimal reading requirements, clear voice prompts and repeatable instructions.
- Cartoon characters, gentle motion, sound effects and cheerful feedback without overwhelming the child.
- English learning content, with Malayalam voice guidance as an optional enhancement.
- Offline access to core lessons; no ads, unnecessary tracking or required account creation in the MVP.

2. Student onboarding and class selection

First launch flow: animated splash screen -> welcome -> student name -> class selection -> avatar selection -> home dashboard. Save the profile locally to avoid repeated setup.

Field	MVP behavior
Student name	Parent enters a name; show friendly personalized greetings.
Class	Pre-KG, LKG or UKG.
Avatar	Choose a cartoon friend (e.g., panda, bunny, lion, frog).
Persistence	Store student details and progress on-device.
Future expansion	Multiple profiles, grades 1-2 and higher, optional sync.

3. Class-wise learning scope

Learning area	Pre-KG	LKG	UKG
Alphabet	A-Z recognition	Letters + phonics	Words + early reading
Numbers	1-20	1-50	1-100
Counting	Objects 1-10	Objects 1-50	Objects 1-100
Shapes	Basic shapes	Shape matching	Patterns
Colors	Common colors	Identify colors	Color combinations
Activities	Tap / identify	Match / trace	Puzzles / simple quizzes
Math	Count objects	More / less; add	Basic add / subtract

These are suggested difficulty presets, not a fixed school syllabus. Parents should be able to adjust difficulty and replay lessons.

4. Number learning: 1-100 across 10 levels

Ten levels introduce 10 numbers each: 1-10, 11-20, 21-30, 31-40, 41-50, 51-60, 61-70, 71-80, 81-90 and 91-100. Pre-KG and LKG initially see class-appropriate subsets, while the lesson engine supports the full range.

Reusable number-level activity sequence

- Meet the number: large animated numeral and spoken pronunciation; tap to replay.
- Count with me: animated apples, balloons or animals appear one by one as the narrator counts.
- Trace the number: dotted path, finger drawing and optional animated stroke guidance.
- Find the number: select the correct numeral from large picture-based choices.
- Celebrate: earn a star, unlock the next level, or practice again.

5. Alphabet learning: A-Z in 3 steps

Step	Learning goal	Example interaction
1. Discover	Recognize uppercase/lowercase letters and sounds.	A appears; an apple bounces; voice says "A for Apple."
2. Practice	Trace letters, match objects and practice basic phonics.	Finger traces B; tap the bouncing ball.
3. Play & master	Picture quizzes, missing letters and simple word identification.	Select the object starting with C and earn stars.

6. Additional learning worlds

- Shapes World: drag and match circles, squares, triangles and stars.
- Color Magic: tap colors, paint objects and complete a rainbow.
- Animal Friends: recognize animal names and listen to sounds.
- Puzzle Playground: match fruits, silhouettes, patterns and objects.
- Later versions: simple stories, rhymes, phonics sequences and age-appropriate math games.

7. Engagement and rewards

Use encouraging micro-rewards: animated stars, happy mascot reactions, confetti, collectible stickers and a simple progress garden. A child can repeat any completed activity. Wrong answers trigger a helpful hint rather than punishment. Avoid pressure timers, loss mechanics and addictive streaks.

8. Screen map and navigation

Splash -> Student Setup -> Kids Home -> Choose Learning World -> Select Level -> Play Activity -> Reward -> Continue / Replay. A separate Parent Zone is accessed through an adult-oriented gate.

Screen	Main elements
Splash	App logo, mascot animation, short welcome sound.
Onboarding	Name, class cards, avatar selection, continue button.
Home	Large alphabet, numbers, shapes, colors and games tiles.
Lesson	Full-screen visuals, narration, touch interaction, repeat and exit.
Reward	Stars, celebration, next lesson and replay.
Parent Zone	Profile, difficulty, audio/language, progress and reset controls.

9. Flutter technology and architecture

Area	Recommendation	Purpose
App framework	Flutter / Dart	Single Android and iOS codebase
State management	Riverpod	Lesson and app state
Routing	GoRouter	Screen navigation
Animations	Rive + Lottie	Interactive mascot and celebrations
Mini-games	Flutter / optional Flame	Tap, drag, match, game loops
Local persistence	Drift or Isar	Student profiles and progress
Audio	just_audio	Narration and effects
Tracing	CustomPainter	Finger-based letter and number paths
Assets	SVG / WebP / Rive	Optimized visual assets

Suggested modular code structure

lib/app (theme, router), lib/core (audio, animation, database, widgets), lib/features (onboarding, student_profile, home, alphabet, numbers, tracing, mini_games, rewards, parent_zone), lib/shared/models; assets/characters, animations, audio, illustrations and lessons.

Data-driven lessons: Define lesson IDs, objects, audio, animation paths, activities, class eligibility and unlock criteria in JSON or equivalent structured content. Reuse one lesson engine instead of creating 126 separate screens.

10. Delivery phases and timeline

Period	Phase	Deliverables
Weeks 1-2	UX and design	Mascot, visual system, screen designs, animation prototype
Week 3	Foundation	App shell, onboarding, profiles, local persistence
Weeks 4-6	Learning engine	Alphabet, numbers, audio, tracing, reusable activities
Weeks 7-8	Games + rewards	Quizzes, progress, achievements, parent controls
Week 9	QA + optimization	Device tests, offline tests, performance, child usability
Week 10	Release	Store assets, privacy materials, beta tests and submissions

Estimate: roughly 8-10 weeks for a focused MVP with an experienced Flutter developer and available assets. Fully bespoke animation, multilingual narration and illustration production may extend this schedule.

11. MVP scope and future roadmap

Version 1 - launch scope

- Pre-KG, LKG and UKG profiles; student name and avatar.
- A-Z alphabet in 3 learning steps and 1-100 number engine in 10 levels.
- Animated visual lessons, English narration, tracing and simple matching/counting quizzes.
- Rewards, local progress, offline lessons and a basic Parent Zone.
- Android and iOS builds with child-safe UX and app-store preparation.

Future releases

- Malayalam narration and more languages; additional shapes, colors, animals and stories.
- Grades 1-2 and higher, richer math and phonics progression.
- Multiple student profiles, optional parent sync and school/teacher dashboard.
- More interactive games and content packs after child usability validation.

12. Child safety, privacy and quality

- Keep personal data minimal and preferably on-device; document retention and deletion behavior.
- Avoid advertising and external links inside the child experience; protect sensitive controls with a parental gate.
- Use accessible contrast, large controls, clear audio and non-timed alternatives to difficult gestures.
- Test on entry-level Android devices and recent iPhones; optimize asset sizes and memory use.
- Review the latest Google Play Families and Apple Kids Category requirements before submission.

13. Recommended build order and acceptance criteria

Build order: (1) Finalize designs; (2) create the Flutter foundation; (3) fully implement one number lesson, including animation, narration, tracing, quiz and reward; (4) generalize the engine; (5) populate A-Z and 1-100; (6) test and publish.

Initial MVP acceptance checks

- A new child can create a profile and reach the first activity without reading lengthy instructions.
- Core lessons run with the device offline and progress survives app restarts.
- Each alphabet step and number level follows the same reliable interaction pattern.
- Audio can be repeated or muted; incorrect answers receive supportive feedback.
- Navigation and animations remain responsive on supported Android and iOS devices.
- Parent settings cannot be opened accidentally through a single child tap.

14. Design decisions to finalize before implementation

- Art direction: rounded 3D-style cartoons, flat 2D illustrations or pastel storybook.
- Mascot character and brand name.
- Language for the MVP: English only or English with Malayalam guidance.
- Portrait-only layout or tablet support; device performance targets.
- Exact unlock rules and curriculum depth for each class.

Recommended starting point: Rounded, expressive cartoon characters; bright but balanced colors; English lessons with optional Malayalam guidance; and a reusable touch-based lesson engine. Prioritize one excellent learning interaction before scaling content.